

PENETRATION TESTING LAB -CAT(L

Done by : Nori karthikey;

Reg no: 22BCI0052

QUESTIONS

1. Detect the service/daemon versions for the IP address of your machine using Nmap. What command is used to trace the route from the scanning machine to a target host? Also, list the geographical locations of each hop.

2. How will you capture the DNS packets that Web Surfing generates? (HINT: Use the ipconfig command to empty the DNS cache, then empty your browser cache, Open Wireshark and filter it to obtain your IP address, Start packet capture, and Visit <http://www.ietf.org>, Stop the capture)

Answer the following questions:

- **To what IP address is the DNS query message sent? Use ipconfig to determine the IP address of your local DNS server. Are these two IP addresses the same?**
- **Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?**

3. What does a Ping sweep do? How will you run a ping sweep without using the Nmap tool? Display the output as a screenshot

Solutions:

1. Detect the service/daemon versions for the IP address of your machine using Nmap. What command is used to trace the route from the scanning machine to a target host? Also, list the geographical locations of each hop.

The NMAP command for the service versions is :

```
sudo nmap -sV -F -oN quick_nmap_192.168.1.4.txt 192.168.1.4
```

```

(kali@vbox)-[~]
$ hostname -I
192.168.1.5

(kali@vbox)-[~]
$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ac:26:d2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.5/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
        valid_lft 536sec preferred_lft 536sec
    inet6 fe80::a00:27ff:feac:26d2/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@vbox)-[~]
$

```

```

(kali@vbox)-[~]
$ nmap -sV -f -oN quick_nmap_192.168.1.4txt 192.168.1.4
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-06 07:47 EST
Stats: 0:00:06 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 47.83% done; ETC: 07:47 (0:00:07 remaining)
Stats: 0:00:11 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 86.96% done; ETC: 07:47 (0:00:02 remaining)
Nmap scan report for 192.168.1.4
Host is up (0.0036s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8089/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:53:7F:A5 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.29 seconds

```

Many services are outdated/insecure (old OpenSSH, Apache 2.2.8, BIND 9.4.2, vsftpd 2.3.4) and the deliberate bind shell on port 1524 confirms this is a purposely vulnerable Metasploitable2 VM. These services represent clear attack surfaces.

Running traceroute:

traceroute -n 192.168.1.4

This is the technique used finding the no of hops it takes in between the communication

```
File Actions Edit View Help
(kali@vbox)-[~]
$ traceroute -n 192.168.1.4
traceroute to 192.168.1.4 (192.168.1.4), 30 hops max, 60 byte packets
 1  192.168.1.4  5.141 ms  5.037 ms  5.026 ms
```

The scan confirms 192.168.1.4 (Metasploitable2) is up with multiple outdated services and known vulnerable points (e.g., vsftpd, bind shell). Traceroute shows the host is local, so it is reachable for further lab work.

Running fping (ping sweep):

fping -a -g 192.168.1.0/24 2>/dev/null | tee fping_alive.txt

```
(kali@vbox)-[~]
$ traceroute -n 192.168.1.4
traceroute to 192.168.1.4 (192.168.1.4), 30 hops max, 60 byte packets
 1  192.168.1.4  5.141 ms  5.037 ms  5.026 ms

(kali@vbox)-[~]
$ sudo apt install -y fping
[sudo] password for kali:
fping is already the newest version (5.1-1).
fping set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1376
192.168.1.1
192.168.1.2
192.168.1.3
192.168.1.4
192.168.1.5
```

The traceroute command returned a single hop to the target 192.168.1.4 with round-trip times

about 5 ms. This shows the target is on the same local network (host-only / LAN) and is directly reachable with no intermediate public routers.

The fping ping-sweep reported live hosts on the subnet and included 192.168.1.4 (the Metasploitable2 VM). Other addresses such as 192.168.1.1, 192.168.1.2, 192.168.1.3 also responded, indicating multiple active devices on the local network.

2. How will you capture the DNS packets that Web Surfing generates? (HINT: Use the ipconfig command to empty the DNS cache, then empty your browser cache, Open Wireshark and filter it to obtain your IP address, Start packet capture, and Visit <http://www.ietf.org>, Stop the capture)

Answer the following questions:

- To what IP address is the DNS query message sent? Use ipconfig to determine the IP address of your local DNS server. Are these two IP addresses the same?**
- Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”?**

```
C:\Users\karth>ipconfig /flushdns
```

```
Windows IP Configuration
```

```
Successfully flushed the DNS Resolver Cache.
```

```
C:\Users\karth>ipconfig /all
```

Windows IP Configuration

```
Host Name . . . . . : LAPTOP-9S7R04TC
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
```

Ethernet adapter Ethernet 2:

```
Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-05
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
NetBIOS over Tcpip. . . . . : Enabled
```

Unknown adapter McAfee VPN:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : TAP-Windows Adapter V9
Physical Address. . . . . : 00-FF-B7-66-DA-87
```

Wireless LAN adapter Local Area Connection* 1:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 84-1B-77-E3-5A-CC
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Local Area Connection* 2:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : 86-1B-77-E3-5A-CB
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
```

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Wi-Fi 6 AX200 160MHz
Physical Address. . . . . : 66-CC-02-55-B9-D4
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 172.17.210.194(Preferred)
Subnet Mask . . . . . : 255.255.248.0
```

2435 2.446801	172.17.210.194	172.17.208.1	DNS	72 Standard query 0x3090 HTTPS www.ietf.org
2436 2.406386	172.17.210.194	172.17.208.1	DNS	72 Standard query 0x7a6f A www.ietf.org
2487 2.446850	172.17.208.1	172.17.210.194	DNS	117 Standard query response 0x3090 HTTPS www.ietf.org HTTPS
2511 2.467629	172.17.208.1	172.17.210.194	DNS	104 Standard query response 0x7a6f A www.ietf.org A 104.16.44.99 A 104.16.45.99
2692 2.608434	172.17.210.194	172.17.208.1	DNS	75 Standard query 0xfa86 HTTPS static.ietf.org
2693 2.608818	172.17.210.194	172.17.208.1	DNS	75 Standard query 0x6963 A static.ietf.org
2768 2.658031	172.17.208.1	172.17.210.194	DNS	120 Standard query response 0xfa86 HTTPS static.ietf.org HTTPS
2785 2.668640	172.17.208.1	172.17.210.194	DNS	107 Standard query response 0x6963 A static.ietf.org A 104.16.45.99 A 104.16.44.99
3223 2.972744	172.17.210.194	172.17.208.1	DNS	78 Standard query 0x440e HTTPS analytics.ietf.org
3226 2.973143	172.17.210.194	172.17.208.1	DNS	78 Standard query 0x2431 A analytics.ietf.org
3428 3.119522	172.17.208.1	172.17.210.194	DNS	123 Standard query response 0x440e HTTPS analytics.ietf.org HTTPS
3468 3.143881	172.17.208.1	172.17.210.194	DNS	110 Standard query response 0x2431 A analytics.ietf.org A 104.16.44.99 A 104.16.45.99
4039 3.539335	172.17.210.194	172.17.208.1	DNS	82 Standard query 0x5f5b HTTPS growthbook.justdone.ai
4040 3.539823	172.17.210.194	172.17.208.1	DNS	82 Standard query 0xb9c7 A growthbook.justdone.ai
4411 3.800718	172.17.208.1	172.17.210.194	DNS	189 Standard query response 0xb9c7 A growthbook.justdone.ai CNAME d2jokphrbv8pjf.cloudfront.net A 108.157.128.112 A 108.157.128.25 A 108.157.128.112
4475 3.841122	172.17.208.1	172.17.210.194	DNS	147 Standard query response 0x5f5b HTTPS growthbook.justdone.ai CNAME d2jokphrbv8pjf.cloudfront.net HTTPS
7403 7.594864	172.17.210.194	172.17.208.1	DNS	91 Standard query 0x0457 HTTPS content-autofill.googleapis.com
7408 7.595371	172.17.210.194	172.17.208.1	DNS	91 Standard query 0xbb8e A content-autofill.googleapis.com
7456 7.629513	172.17.208.1	172.17.210.194	DNS	347 Standard query response 0xbb8e A content-autofill.googleapis.com A 142.250.183.138 A 142.250.183.170 A 142.250.183.170 A 142.250.206.112
7564 7.711522	172.17.208.1	172.17.210.194	DNS	148 Standard query response 0x0457 HTTPS content-autofill.googleapis.com SOA ns1.google.com
15449 15.542022	172.17.210.194	172.17.208.1	DNS	70 Standard query 0x5521 HTTPS google.com
15450 15.542346	172.17.210.194	172.17.208.1	DNS	70 Standard query 0x0112 A google.com
15470 15.555674	172.17.208.1	172.17.210.194	DNS	167 Standard query response 0x5521 HTTPS google.com HTTPS NS ns4.google.com NS ns3.google.com NS ns2.google.com NS ns1.google.com

2435 2.446801	172.17.210.194	172.17.208.1	DNS	72 Standard query 0x3090 A www.ietf.org
2487 2.446850	172.17.208.1	172.17.210.194	DNS	117 Standard query response 0x3090 HTTPS www.ietf.org HTTPS
2511 2.467629	172.17.208.1	172.17.210.194	DNS	104 Standard query response 0x7a6f A www.ietf.org A 104.16.44.99 A 104.16.45.99
2692 2.608434	172.17.210.194	172.17.208.1	DNS	75 Standard query 0xfa86 HTTPS static.ietf.org
2693 2.608818	172.17.210.194	172.17.208.1	DNS	75 Standard query 0x6963 A static.ietf.org
2768 2.658031	172.17.208.1	172.17.210.194	DNS	120 Standard query response 0xfa86 HTTPS static.ietf.org HTTPS
2785 2.668640	172.17.208.1	172.17.210.194	DNS	107 Standard query response 0x6963 A static.ietf.org A 104.16.45.99 A 104.16.44.99
3223 2.972744	172.17.210.194	172.17.208.1	DNS	78 Standard query 0x440e HTTPS analytics.ietf.org
3226 2.973143	172.17.210.194	172.17.208.1	DNS	78 Standard query 0x2431 A analytics.ietf.org
3428 3.119522	172.17.208.1	172.17.210.194	DNS	123 Standard query response 0x440e HTTPS analytics.ietf.org HTTPS
3468 3.143881	172.17.208.1	172.17.210.194	DNS	110 Standard query response 0x2431 A analytics.ietf.org A 104.16.44.99 A 104.16.45.99
4039 3.539335	172.17.210.194	172.17.208.1	DNS	82 Standard query 0x5f5b HTTPS growthbook.justdone.ai
4040 3.539823	172.17.210.194	172.17.208.1	DNS	82 Standard query 0xb9c7 A growthbook.justdone.ai
4411 3.800718	172.17.208.1	172.17.210.194	DNS	189 Standard query response 0xb9c7 A growthbook.justdone.ai CNAME d2jokphrbv8pjf.cloudfront.net A 108.157.128.112 A 108.157.128.25 A 108.157.128.112
4475 3.841122	172.17.208.1	172.17.210.194	DNS	147 Standard query response 0x5f5b HTTPS growthbook.justdone.ai CNAME d2jokphrbv8pjf.cloudfront.net HTTPS
7403 7.594864	172.17.210.194	172.17.208.1	DNS	91 Standard query 0x0457 HTTPS content-autofill.googleapis.com
7408 7.595371	172.17.210.194	172.17.208.1	DNS	91 Standard query 0xbb8e A content-autofill.googleapis.com
7456 7.629513	172.17.208.1	172.17.210.194	DNS	347 Standard query response 0xbb8e A content-autofill.googleapis.com A 142.250.183.138 A 142.250.183.170 A 142.250.183.170 A 142.250.206.112
7564 7.711522	172.17.208.1	172.17.210.194	DNS	148 Standard query response 0x0457 HTTPS content-autofill.googleapis.com SOA ns1.google.com
15449 15.542022	172.17.210.194	172.17.208.1	DNS	70 Standard query 0x5521 HTTPS google.com
15450 15.542346	172.17.210.194	172.17.208.1	DNS	70 Standard query 0x0112 A google.com
15470 15.555674	172.17.208.1	172.17.210.194	DNS	167 Standard query response 0x5521 HTTPS google.com HTTPS NS ns4.google.com NS ns3.google.com NS ns2.google.com NS ns1.google.com


```

> Frame 2511: 104 bytes on wire (832 bits), 104 bytes captured (832 bits) on 0
> Ethernet II, Src: Intel_b4:b1:dd (00:15:17:b4:b1:dd), Dst: 66:cc:02:55:b9:d4 (66:cc:02:55:b9:d4)
> Internet Protocol Version 4, Src: 172.17.208.1, Dst: 172.17.210.194
> User Datagram Protocol, Src Port: 53, Dst Port: 58773
> Domain Name System (response)
0000 66 cc 02 55 b9 d4 00 15 17 b4 b1 dd 00 00 45 00 f U ..... E
0010 00 5a 3e 50 00 00 40 11 41 5c ac 11 d0 01 ac 11 Z>P @ A\.....
0020 d2 c2 00 35 e5 95 00 46 55 14 7a 6f 81 80 00 01 ..... S... F U z o .....
0030 00 02 00 00 00 00 03 77 77 77 04 69 65 74 66 03 ..... w w w i e t f .....
0040 6f 72 67 00 00 01 00 01 c0 0c 00 01 00 01 00 00 org.....
0050 01 2c 00 04 68 10 2c 63 c0 0c 00 01 00 01 00 00 , h y c .....
0060 01 2c 00 04 68 10 2d 63 , h y c .....

```

The DNS query for www.ietf.org was sent from my machine (172.17.210.194) to the DNS server 172.17.208.1. This matches the DNS server setting from ipconfig. The query type was A (Host Address), requesting the IPv4 address of the domain. The query packet contained no answers, whereas the DNS response contained two IP addresses: 104.16.44.99 and 104.16.45.99, confirming that the domain was successfully resolved.

3. What does a Ping sweep do? How will you run a ping sweep without using the Nmap tool? Display the output as a screenshot

A **ping sweep** probes a range of IP addresses with ICMP Echo Requests to discover which hosts are reachable on a network. It is a basic host-discovery technique used in network mapping and reconnaissance; responding hosts are considered “alive” and worth further analysis. Ping sweeps can be done with specialized tools (fping) for speed or with simple shell loops when tools are unavailable

```
(kali@vbox)-[~]  
$ sudo fping -a -g 192.168.1.0/24 2>/dev/null | tee ping_results_192.168.1.0-24.txt  
192.168.1.1  
192.168.1.2  
192.168.1.3  
192.168.1.4  
192.168.1.5
```

sudo apt update && sudo apt install -y fping (for installing which is done in question 1)

sudo fping -a -g 192.168.1.0/24 2>/dev/null | tee ping_results_192.168.1.0-24.txt